AI-ENABLED CYBERATTACK DETECTION

Adversarially Robust Federated Intrusion Detection

Vinit Soni (E23CSEU0437) · Final submission · November 2026

46.8 M
traffic windows audited

4
threat models evaluated

10
federated clients · Azure + laptop

76
tests, all numbers scripted

Five research questions, four threat models

RQ1 — **Detection** — How well do an MLP and tree models detect CICIoT2023 attacks under a leakage-aware split — and at what false-positive cost?

RQ2 — **Federation** — What does federated training (FedAvg / FedProx, IID → strongly skewed clients) cost versus centralized — in a simulator and in a real Flower deployment?

RQ3 — **Poisoning** — How much damage do malicious clients do to the global model, and do robust aggregators (median, trimmed mean, norm clip) hold?

RQ4 — **Evasion** — Can a black-box attacker that may only pad, slow down and re-TTL its own packets make detected attacks look benign?

RQ5 — **LLM-planned attacks** — On GridStratLLM, do network signatures of LLM-coordinated smart-grid attacks transfer to a held-out planning LLM?

The dataset fights back: hygiene before modelling

CICIOT2023 · 46.8M WINDOWS · 39 FEATURES · 34 ATTACK CLASSES

55.0 %

of rows are exact duplicates — de-duplicated within splits

30.6 %

share their feature vector with another class (DDoS vs DoS)

6 feats

encode the extraction window — a shortcut, excluded

- Leakage-aware split: file-level where possible, purged time blocks otherwise; 12.2 % of test vectors still occur in train — reported separately as “unseen”.
- **Result: honest macro-F1 of 0.56–0.75 — far below the >0.99 common in the literature, by design, not by failure.**
- GridStratLLM (smart grid): BagIt-verified, 436 attack sessions planned by 4 LLMs + 1.75 h normal operation.

Every number in report & slides is generated by a script in the repo — nothing copied from papers.

RQ1 · The operating point matters more than the model

CENTRALIZED DETECTION, 8-CLASS TASK, HELD-OUT TEST

19–29 %

of benign windows flagged at plain argmax — undeployable

1.0–1.3 %

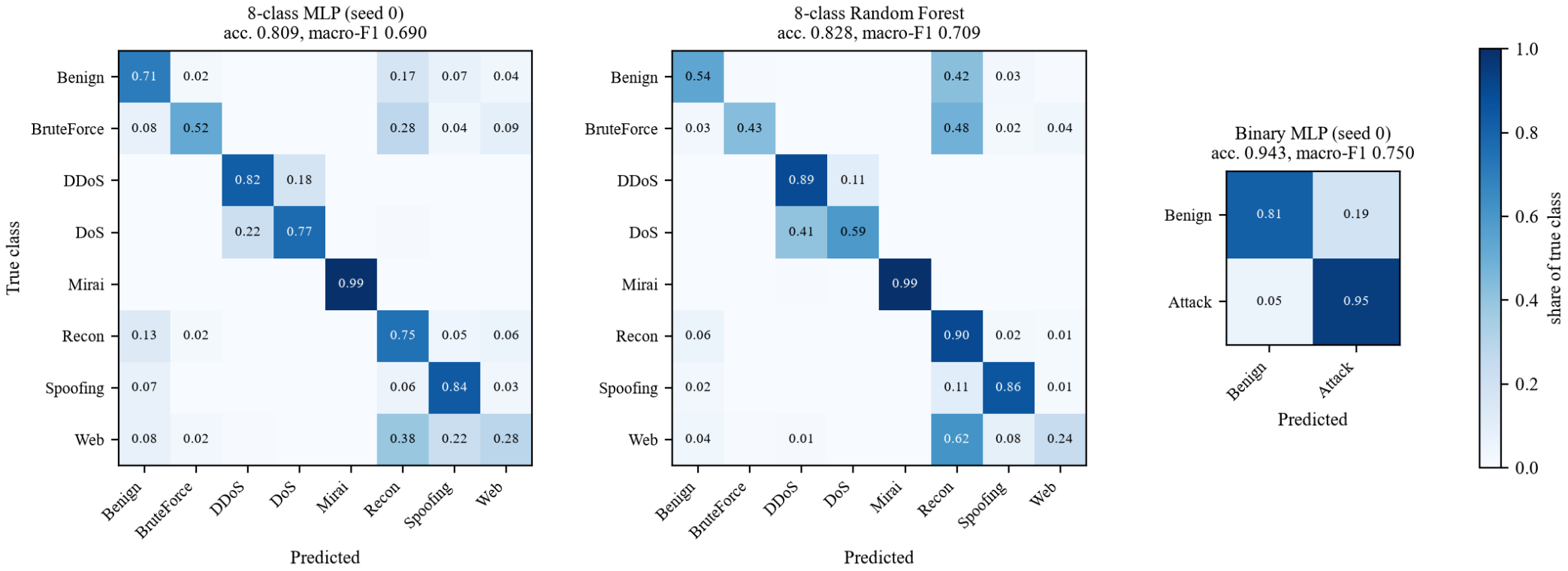
false alarms with a threshold calibrated on validation

83–86 %

of attack windows still detected at the 1 % FPR rule

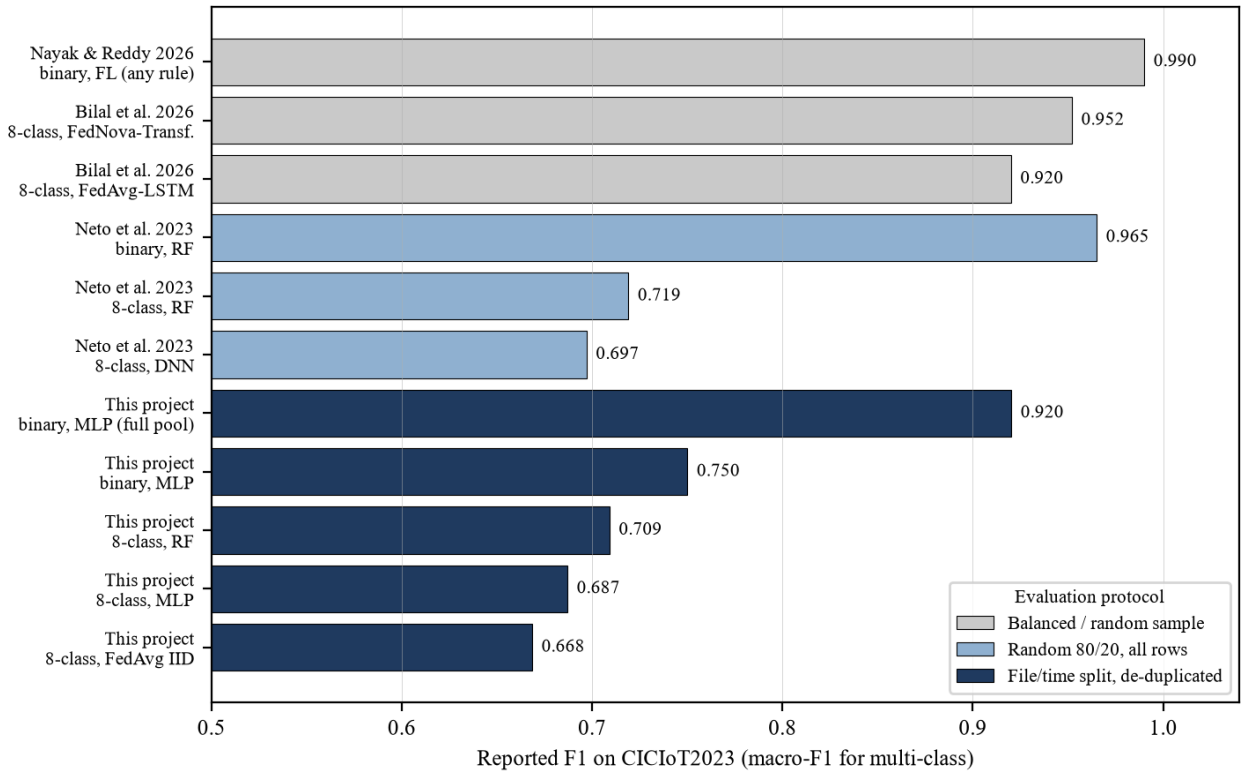
- Calibration transfers: the 1 % validation target achieves 0.7–1.3 % on test, across all models and tasks.
- Tuned v2 MLP gains only 0.3–0.7 detection points over v1 — HistGradientBoosting is the best detector at 1 % FPR (binary +2.8 points), training in seconds.
- The Random Forest “wins” macro-F1 only by flagging 2–3× more benign traffic — model comparisons are meaningless without a fixed operating point.

RQ1 · Accuracy flatters; the confusion matrix shows why



- 8-class accuracy 0.807 (MLP) / 0.828 (RF), but macro-F1 only 0.687 / 0.709. Binary MLP: accuracy 0.943, 0.990 on the full test pool.
- **Errors sit where classes really overlap: DDoS ↔ DoS (18–22 %), Web and Brute force → Recon, Benign → Recon. Mirai is 0.99.**

Against published work: same results, stricter protocol



Reported CICIoT2023 F1 by evaluation protocol

Study	Task · protocol	F1
Neto 2023	8-class · random 80/20	0.719 / 0.697
Bilal 2026	8-class FL · balanced test	0.920–0.952
Nayak & Reddy 2026	binary FL · balanced	0.989–0.990
This project	**8-class · file/time, de-dup.**	**0.687 / 0.709**
This project	**8-class FedAvg IID**	**0.668**

- 8-class result within 0.01–0.03 of the dataset authors' own benchmark, despite a stricter split.
- The 0.92–0.99 scores come from balanced or random protocols that hide benign traffic.
- **None of the four studies tests evasion, a fixed FPR or LLM-planned attacks — this project tests all three.**

Real traffic: the lab result does not transfer

10 MIN OF A LAPTOP ON HOME WI-FI · NO ATTACKS · EVERY ALARM IS FALSE

86–89 %

of windows flagged by the 1 % rule (1.2 % on the CICIoT2023 test set)

98–99 %

of QUIC / HTTP-3 windows flagged — 64 % of the traffic, absent from the dataset

53–59 %

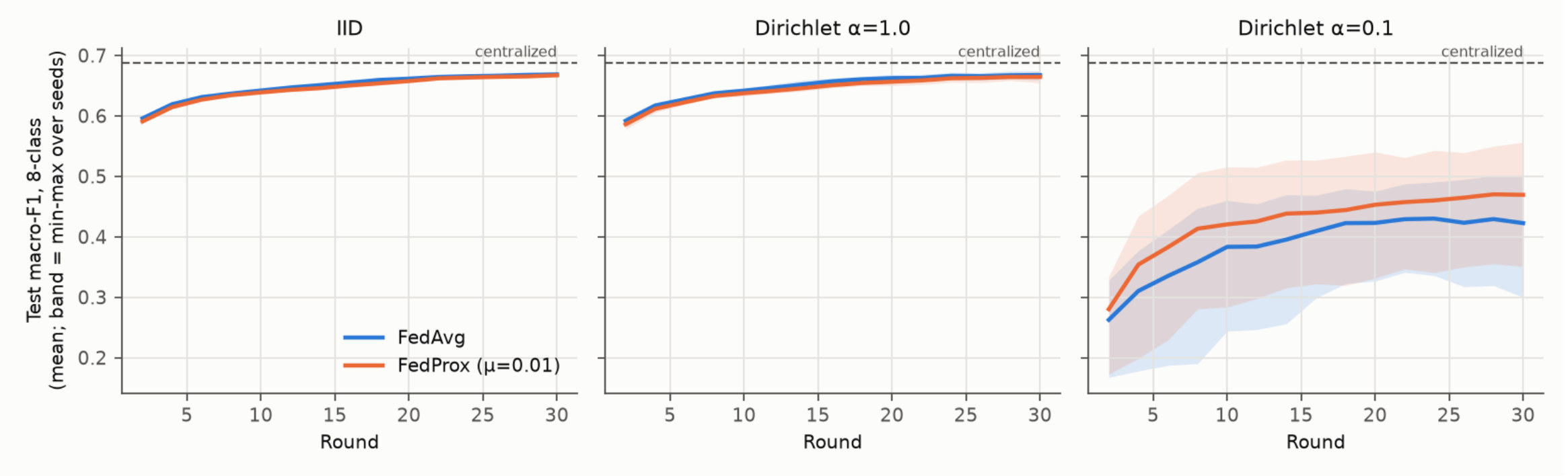
attack detection left after recalibrating on local traffic (was 85 %)

Traffic (no attacks)	Frozen	Hardened
CICIoT2023 test, benign	1.2 %	1.2 %
Laptop, home Wi-Fi	88.6 %	85.5 %
of which TCP browsing	70.0 %	62.5 %
Smart grid, normal operation	99.9 %	35.5 %

False-alarm rate at the 1 % rule

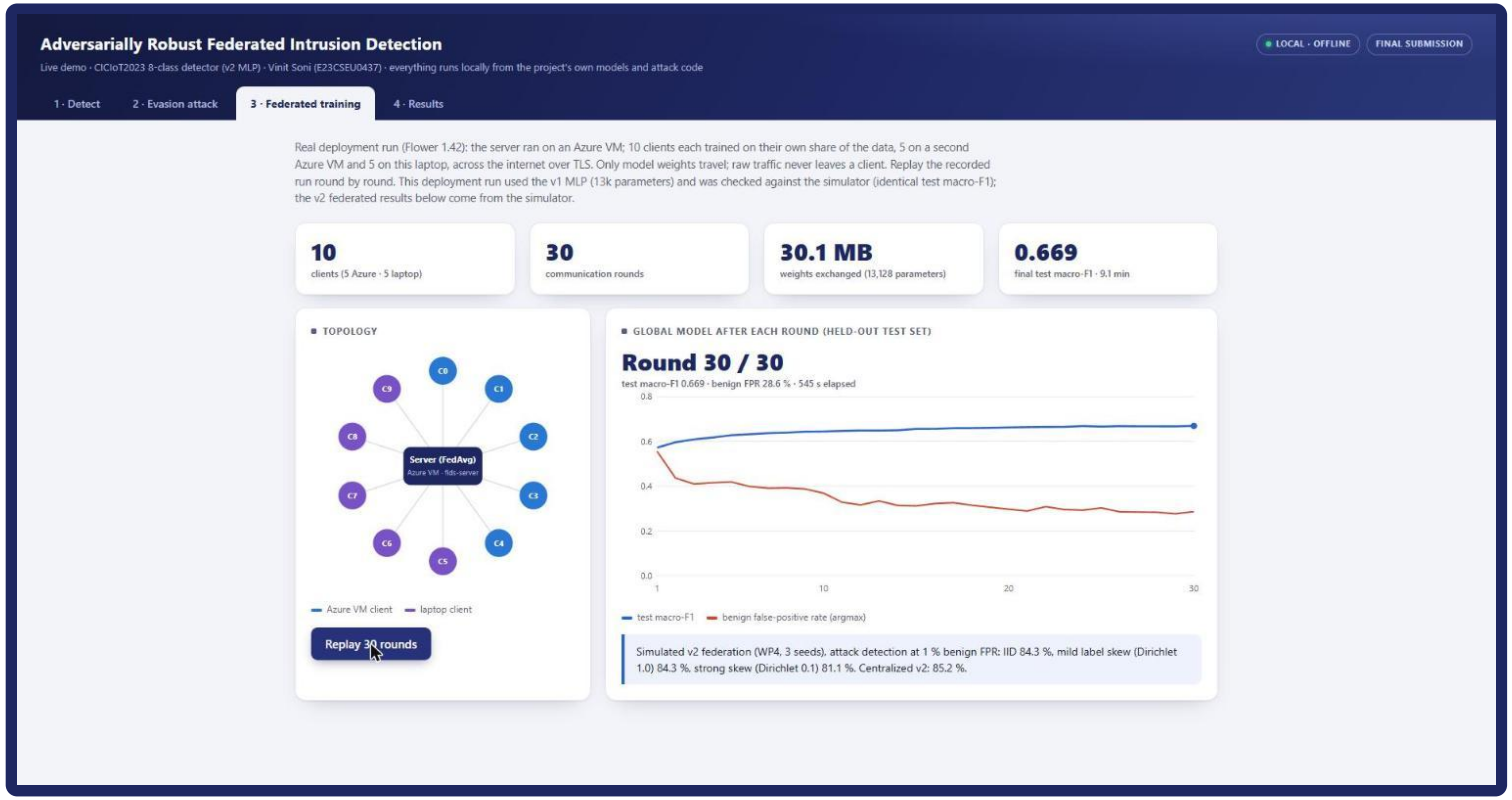
- Not a converter bug: features checked against the dataset's benign rows; Windows capture timing explains only ~4 points.
- A threshold cannot fix a feature-space gap — it sits within 10^{-8} of 1 and misses a third of the attacks.
- **Deployment needs the network's own benign traffic in training — exactly what federated clients on those networks provide.**

RQ2 · Federation is cheap — until the skew is strong



- IID / mild skew (α = 1.0): federated detection at 1 % FPR is only 0.6–2.5 points below centralized after 30 rounds (~30–105 MB traffic).
- 2 local epochs per round is the best realism trade: same traffic, up to +1.4 points, converges in half the rounds.
- **Strong label skew (α = 0.1) breaks training in every task — macro-F1 −0.12–0.26, FPR doubles. Neither FedProx nor 60 instead of 30 rounds fixes it.**

The simulator survives contact with reality

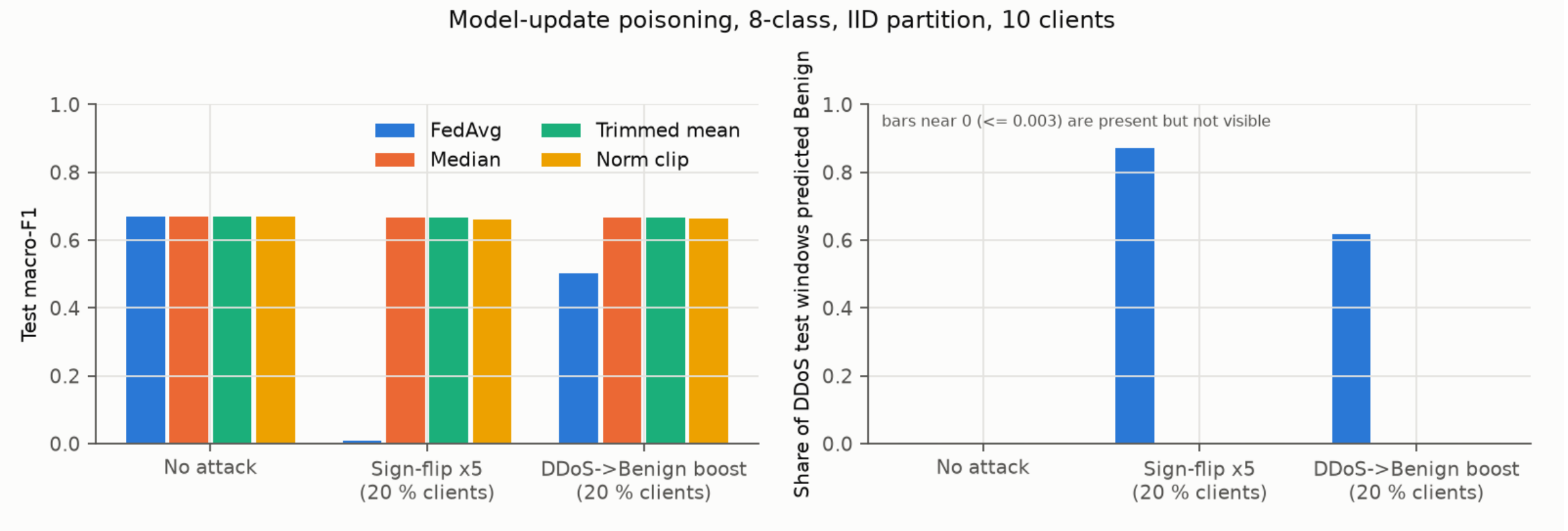


Dashboard replay of the recorded run (demo, tab 3)

REAL FLOWER 1.42 RUN · TLS OVER THE INTERNET

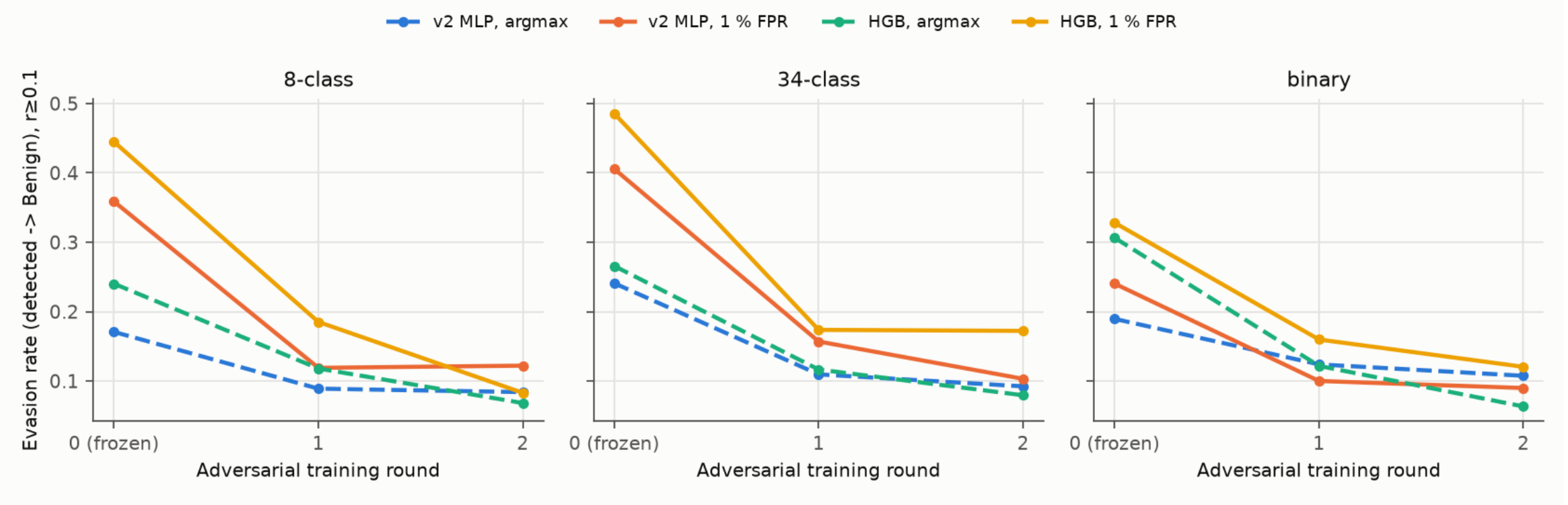
- 1 SuperLink + 5 clients on Azure VMs, 5 clients on this laptop.
- **Test macro-F1 0.6692 — identical to the simulator to four decimals.**
- 30.05 MB exchanged, exactly as estimated; 561 s vs 92 s on GPU.
- Only model weights travel; raw traffic never leaves a client.
- The trained model is served from the Azure VM: HTTPS, API key, SSH tunnel — 400 test windows served identically to offline.

RQ3 · Two bad clients beat FedAvg; medians hold the line



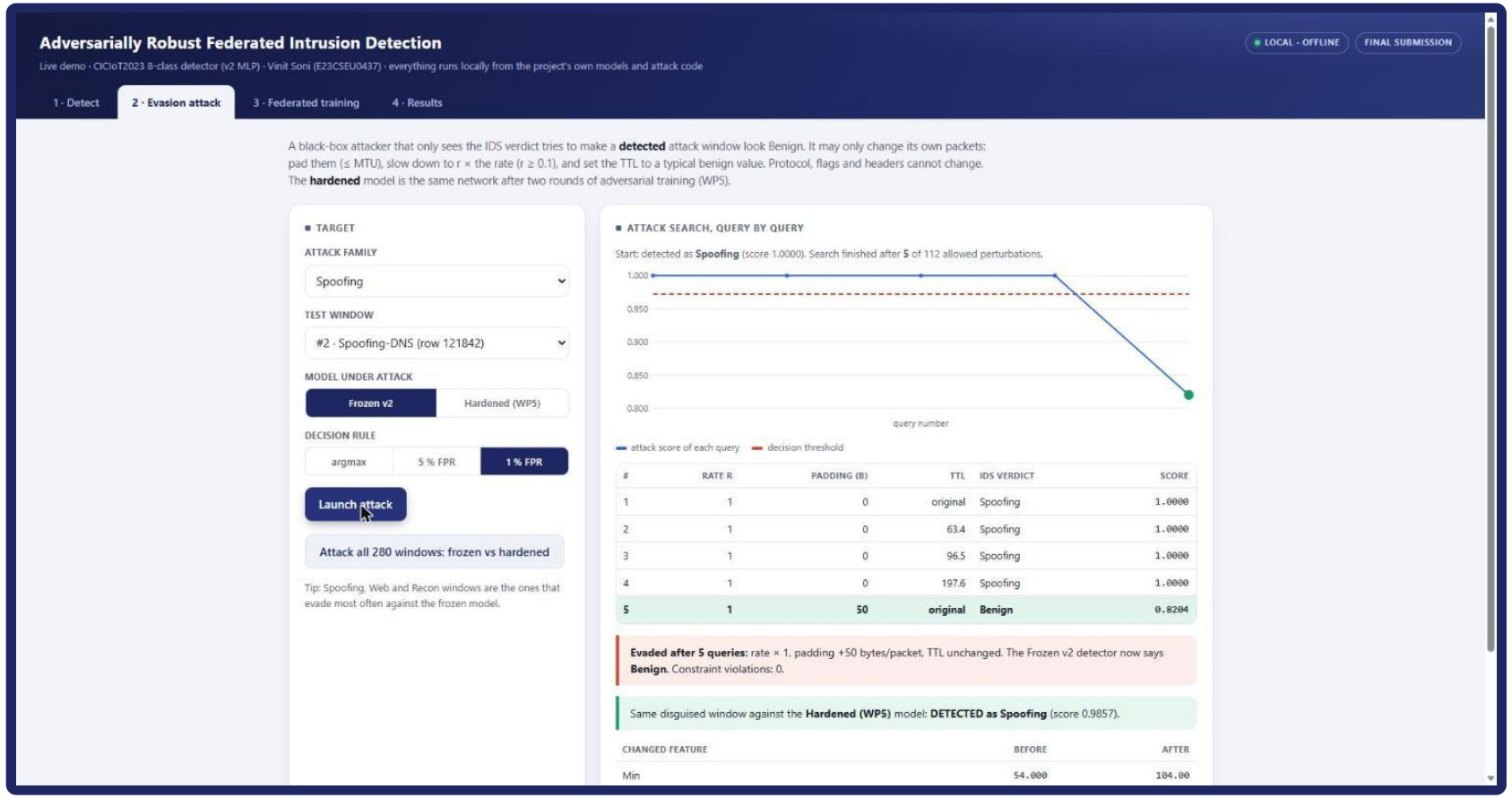
- 2 of 10 malicious: sign-flip collapses FedAvg to macro-F1 ≤ 0.04; targeted poisoning hides up to 62 % of DDoS as benign.
- Median / trimmed mean / norm clip neutralise both attacks at ≤ 0.001 clean cost (IID) — but the trimmed mean fails at 30 % malicious, exactly as its assumption predicts.

RQ4 · The deployable threshold is the most evadable one



- Black-box, label-only attacker (pad · slow · re-TTL, ≤ 112 queries): evades the 1 %-FPR rule for 24–49 % of detected windows — 1.1–2.1× more often than argmax.
- **Two rounds of adversarial training cut evasion to 8–17 % at ≈ zero clean cost; Spoofing & Web — the families that already overlap benign — stay evadable.**

Evaded in 5 queries — the hardened model catches it

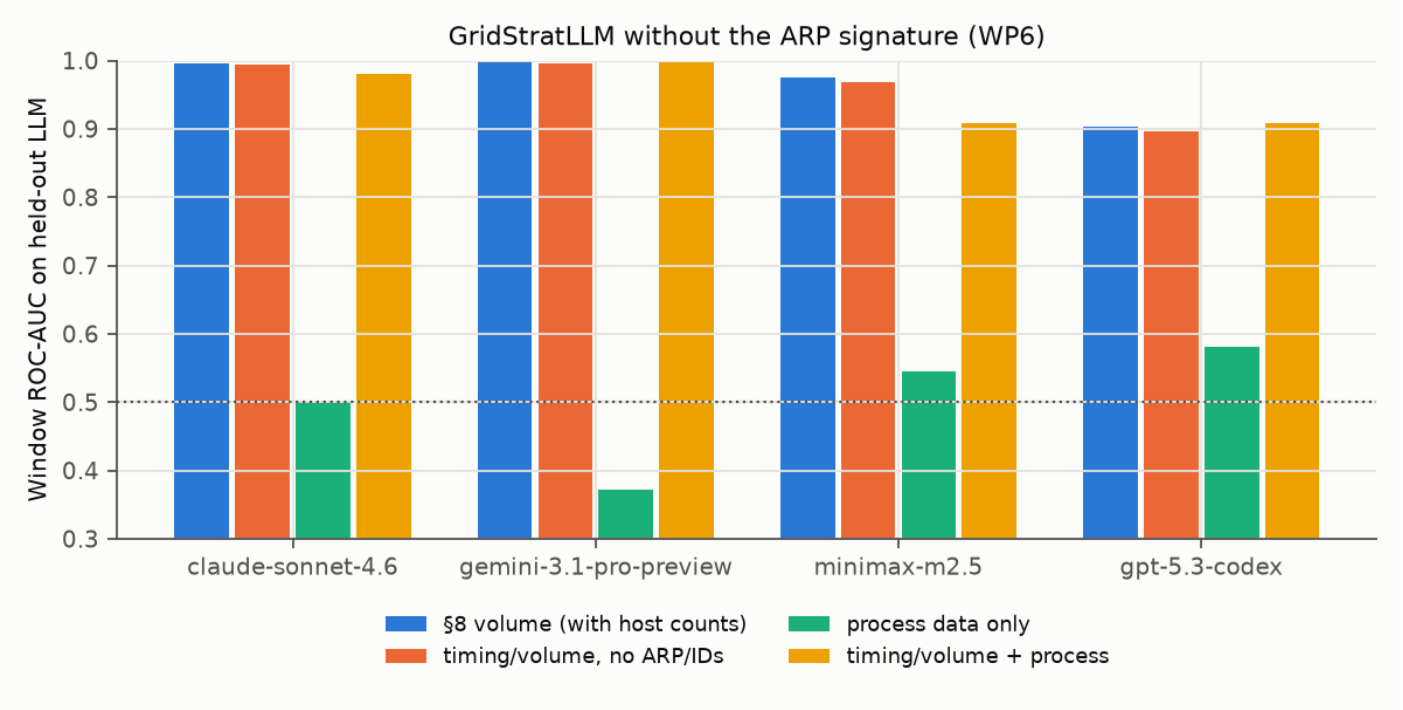


Query-by-query replay in the demo dashboard (tab 2)

SPOOFING-DNS WINDOW, 1 % FPR RULE

- Queries 1–4: every disguise still detected.
- Query 5: +50 B padding per packet → verdict flips to Benign.
- **The same disguised window is still detected as Spoofing by the WP5-hardened model (score 0.986).**
- 0 constraint violations — protocol, flags and headers untouched.

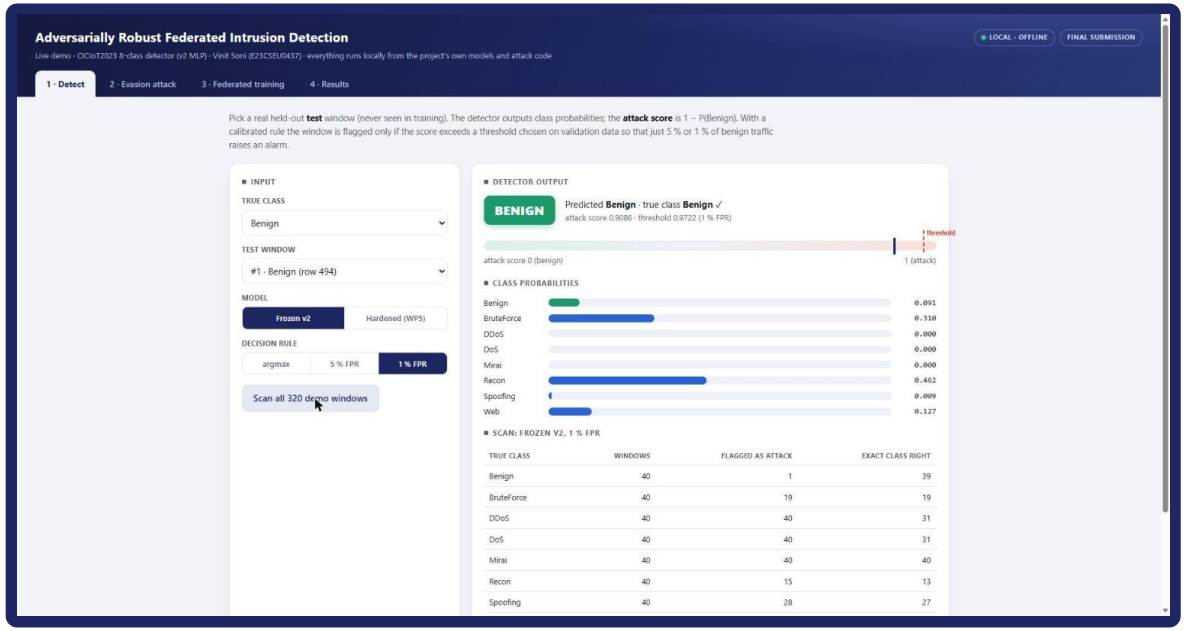
RQ5 · The toolkit betrays the attack, not the LLM



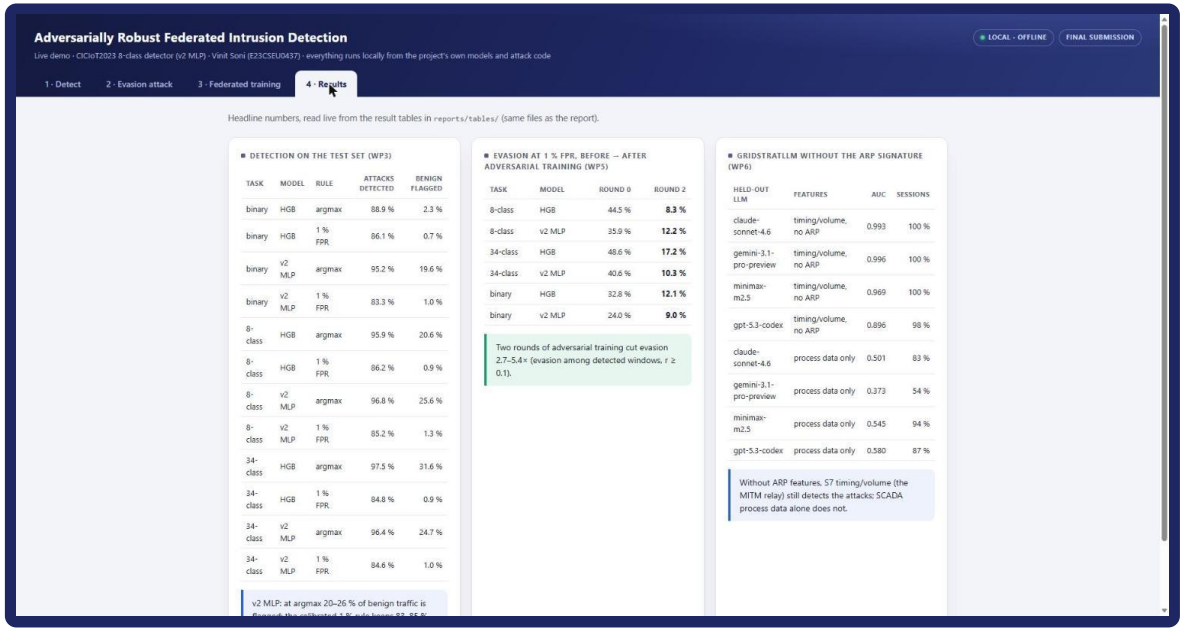
Leave-one-LLM-out AUC without the ARP signature (WP6)

- Signatures transfer across planning LLMs: window AUC 0.90–1.00 on every held-out LLM — all 436 sessions use the same MITM primitives.
- A classical rule (one IP, two MACs) already flags 89–99 % of attack windows with zero false alarms.
- **Remove the ARP features entirely: AUC drops ≤ 0.007 — the MITM relay itself (S7 timing/volume) still gives the attack away.**
- SCADA process data alone carries no usable signal (AUC 0.37–0.58). Attacks without a MITM step remain untestable on this dataset.

Everything shown live, from the project's own artifacts



Detect · pick any held-out window, frozen vs hardened model



Results · headline tables read live from reports/tables/

- Local FastAPI dashboard, no external services: real test windows, the real models, the real attack code (scripts/run_demo.ps1).
- Demo sample at 1 % FPR: frozen model — 64 of 197 detected windows evaded (32.5 %); hardened — 31 of 198 (15.7 %).

What decided the outcome

1. **Data hygiene first** — Duplicates, cross-class conflicts and shortcut features would have inflated every later number. Honest splits set the real baseline.

2. **Fix the operating point** — Argmax comparisons are meaningless; a calibrated 1 %-FPR rule makes models deployable — and comparable.

3. **Robustness is assumptions** — Median holds to 30 % malicious; trimmed mean fails past its trim; adversarial training beats only the attacker it saw.

4. **Federation works, skew doesn’t** — Real deployment matches simulation; strong label skew remains the open problem.

5. **Tools, not intelligence** — LLM-planned grid attacks are caught by their common toolkit — the planning model is irrelevant to the defender.

REPRODUCIBLE END TO END

Every table, figure and demo number regenerates from the repository

Code & report github.com/vinitsoni03/federated-ids · 76 tests · run records per experiment

Live demo scripts/run_demo.ps1 → localhost dashboard · detect, attack, federate, results

Known limits one dataset · does not transfer to home traffic without local data · α = 0.1 unsolved

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